3.2

ADULTS

Cardiovascular System Infections

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Infective endocarditis			
Empirical treatment – in severely ill patients			
Native valve, **Prosthetic valve >1 year post surgery** *Streptococcus viridans* Other *Streptococci* *Enterococcus spp.* *Staphylococcus spp.*	Benzylpenicillin 4 MU IV q4H (total 24 MU daily) **OR** Ampicillin 12 g/day IV q4H-q6H + Cloxacillin 2 g IV q4H + Gentamicin 3 mg/kg IV q24H	Cefazolin 2 g IV q8H + Gentamicin 1 mg /kg IM **OR** IV q8H **Penicillin Allergy:** Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H + Gentamicin 3 mg/kg IV q24H	• Duration: 4 - 6 weeks • Native valves: 4 weeks • Prosthetic valves: 6 weeks • Gentamicin: 2 - 4 weeks • *Staphylococcus aureus* native valve endocarditis: 6 weeks • Obtain at least 3 sets of blood cultures from different sites before empiric therapy. • Addition of Gentamicin to beta lactam clears culture earlier but does not change mortality. Avoid in elderly and renal impaired. Discontinue once cultures known except for

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			Streptococcal and Enterococcal endocarditis. (Refer below to the Enterococcal section). • For Streptococcal IE, once daily dosing of Gentamicin may be used at 3 mg/kg to reduce renal toxicity (MIC>1.2).
Prosthetic valve endocarditis <1 year post surgery *Streptococcus epidermidis* *Streptococcus viridans* *Streptococcus sp* *Staphylococcus aureus* Rarely, *Enterobacteriaceae,* *Diphtheroid,* Fungi	Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H + Rifampicin** 450 mg PO 12H + Cefepime 2 g IV q8H **OR** Gentamicin 3 mg/kg IV q24H	Daptomycin*** 10 mg/kg q24H + Rifampicin** 450 mg PO 12H + Cefepime 2 g IV q8H **OR** Gentamicin 3 mg/kg IV q24H	• **Rifampicin: To avoid the development of resistance, it should be started after 3-5 days of effective initial Cloxacillin therapy and/or once the bacteraemia has been cleared. • ***To refer to ID for Daptomycin use.
Directed treatment			
Viridans streptococci and *Streptococcus bovis* *MIC estimation needs to be done for these isolates to facilitate management			

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Native and Prosthetic Valves			
MIC: < 0.125µg/mL **(Penicillin-susceptible Streptococcus Viridans)**	Benzylpenicillin 3 MU IV q4-6H	Ampicillin 2 g IV q4H **OR** Ceftriaxone 2 g IV q24H **Penicillin Allergy:** Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H	• Duration: • **Native valves:** 4 weeks • **Prosthetic valves:** at least 6 weeks • For Penicillin-susceptible viridans streptococci, monotherapy with Benzylpenicillin, Ampicillin or Ceftriaxone is adequate. • In non-complicated native valve endocarditis with normal renal function, Benzylpenicillin or Ceftriaxone with Gentamicin for a duration of 2 weeks can be considered.
MIC: > 0.125µg/mL-2µg/mL **(Penicillin-relatively resistant Streptococcus Viridans)**	Benzylpenicillin 4 MU IV q4H (total 24 MU daily) **OR** Benzylpenicillin 24 MU IV as continuous infusion + Gentamicin* 3 mg/kg IV q24H	Ceftriaxone 2 g IV q24H + Gentamicin* 3 mg/kg IV q24H **Penicillin Allergy:** Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H	• *For Penicillin-relatively resistant Streptococcus viridans, Gentamicin has to be added to the regime. • Duration of Gentamicin: 2 weeks for both native and prosthetic valve IE. • TDM Gentamicin: aim for pre-dose (trough) serum level of < 1mg/L. • After 10–14 days of IV therapy, IV OPAT or outpatient oral antibiotic therapy can be considered if the patient is clinically stable. Decision for drug regime is to be made by ID and cardiology team.

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
		+ Gentamicin* 3 mg/kg IV q24H	
MIC > 2µg/mL **(Penicillin-resistant Streptococcus Viridans & Streptococcus bovis)**	Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H + Gentamicin 3 mg/kg IV q24H		
Enterococcus * For Enterococcal IE, please check with the lab for HLAR (High Level Aminoglycoside Resistance). If present, please stop Gentamicin.			
Native and Prosthetic Valves			
Sensitive to Penicillin & Gentamicin **(MIC penicillin <8 mg/L)**	Ampicillin 2 g IV q4H + Gentamicin 3 mg/kg IV 24H		• **Duration for native valve endocarditis and symptoms <3 months:** 4 weeks of Ampicillin. • **Duration for prosthetic valve endocarditis or symptoms >3 months:** 6 weeks of Ampicillin. • Duration of Gentamicin: 2 weeks for both native and

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			prosthetic valves IE. • After 10–14 days of IV therapy, IV OPAT or outpatient oral antibiotic therapy can be considered if the patient is clinically stable. Decision for drug regime is to be made by ID and cardiology team.
Sensitive to Penicillin but HLAR to Gentamicin, **renal impairment and elderly**	Ampicillin 2 g IV q4H + Ceftriaxone 2 g IV q12H		• Duration: 6 weeks
Resistant to Penicillin, **susceptible to Aminoglycosides and Vancomycin**	Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H + Gentamicin* 3 mg/kg IV q24H		• Duration: at least 6 weeks of Vancomycin and 2 weeks of Gentamicin. • *Gentamicin: aim for pre-dose (trough) serum level of <1mg/L.
Resistant to Penicillin and	Daptomycin* 10 – 12		• Refer for formal ID team consult.

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Vancomycin	mg/kg/day IV q24H + Ampicillin 300 mg/kg/day IV q4-6H		• *Refer ID for Daptomycin use. • Duration: at least 8 weeks.
Staphylococcus aureus			
Native valves Methicillin-sensitive *Staphylococci* (MSSA)	**Left sided endocarditis or complicated right sided endocarditis:** Cloxacillin 2 g IV q4H for 4 to 6 weeks **Right sided endocarditis (tricuspid valve) or uncomplicated endocarditis:** Cloxacillin 2 g IV in q4H for* 2 to 4 weeks	**Penicillin Allergy:** **<u>Immediate type hypersensitivity to penicillin (anaphylaxis):</u>** Daptomycin** 10 mg/kg IV q24H for 4 to 6 weeks **<u>For non-immediate type hypersensitivity:</u>** Cefazolin 2 g IV q8H for 4 to 6 weeks	• *In right sided endocarditis, 2 weeks' regime is sufficient provided the patient fulfils all the following criteria (uncomplicated IE): • MSSA • Absence of associated prosthetic valve or left sided valve infection. • Good response to treatment • Absence of metastatic sites of infection or empyema. • Absence of cardiac and extracardiac complications • < 20 mm vegetation • Absence of severe immunosuppression (<200 CD4 cells/ml) with or without Acquired Immune Deficiency Syndrome (AIDS)

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			• **To refer to ID for Daptomycin use.
Prosthetic valves Methicillin-sensitive *Staphylococci* (MSSA)	Cloxacillin 2 g IV in q4H for ≥ 6 weeks + Gentamicin 3 mg/kg IV q24H for 2 weeks + Rifampicin* 300-450 mg PO q12H for ≥ 6 weeks	**Penicillin Allergy:** **<u>Immediate type hypersensitivity to penicillin (anaphylaxis):</u>** Daptomycin** 10 mg/kg IV q24H for 4 to 6 weeks + Gentamicin 3 mg/kg IV q24H for 2 weeks + Rifampicin* 300-450 mg PO q12H for ≥ 6 weeks **<u>For non-immediate type hypersensitivity:</u>** Cefazolin 2 g IV q8H for 4 to 6 weeks + Gentamicin 3 mg/kg IV	• *Rifampicin: To avoid the development of resistance, it should be started after 3-5 days of effective initial Cloxacillin therapy and/or once the bacteraemia has been cleared. • After 10–14 days of IV therapy, IV OPAT or outpatient oral antibiotic therapy can be considered if the patient is clinically stable. Decision for drug regime is to be made by ID and cardiology team. • **To refer to ID for Daptomycin use.

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
		q24H for 2 weeks + Rifampicin* 300-450 mg PO q12H for ≥ 6 weeks	
Native valves Methicillin-resistant *Staphylococci* (MRSA) **MIC of Vancomycin < 1 mg/mL** **MIC of Vancomycin > 1 mg/mL**	Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H, for 4 to 6 weeks Daptomycin* 10 mg/kg IV q24H for 4 to 6 weeks	Daptomycin* 10 mg/kg IV q24H for 4 to 6 weeks	• *To refer to ID for Daptomycin use.
Prosthetic valves Methicillin-resistant *Staphylococci* (MRSA)	Vancomycin 20 mg/kg (max 2 g) IV single dose, then 15 mg/kg (max 1 g) IV q8-12H, for ≥ 6 weeks		• *Rifampicin: To avoid the development of resistance, it should be started after 3-5 days of effective initial Vancomycin therapy and/or once the bacteraemia has been cleared.

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	+ Gentamicin 3 mg/kg IV q24H for 2 weeks + Rifampicin* 450-600 mg PO q12H for ≥ 6 weeks		
Treatment for culture-negative endocarditis – consultation with an infectious disease specialist is needed			
HACEK, Brucella, Bartonella, Coxiella			
Native and Prosthetic Valves			
HACEK organisms *Haemophilus parainfluenzae, aggregatibacter (*formerly *Haemophilus) Aphrophilus, Cardiobacterium hominis, Eikenella corrodens, Kingella species*	Ceftriaxone 2 g IV q24H	Ampicillin 2 g IV q4H + Gentamicin 3 mg/kg IV q24H for 2 weeks **OR** Ciprofloxacin 400 mg IV **OR** Ciprofloxacin 500 mg PO q12H	• Duration: • Native valves: 4 weeks. • Prosthetic valves: 6 weeks.

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Brucella spp.	Doxycycline 100 mg PO q12H + Rifampicin 600-900 mg PO q24H + Gentamicin 5 mg/kg IV q24H (for first 4 weeks) **OR** Streptomycin 1 g IM q24H	Doxycycline 100 mg PO q12H + Rifampicin 600-900 mg PO q24H + Co-trimoxazole 960 mg PO q12H	• Duration: 3 to 6 months. • Treatment success is defined as Brucella IgG titre < 1:60 (when available). • Referral to Cardiothoracic Surgery (CTS) for surgical resection is indicated. • **Regimen for pregnant patient:** Ceftriaxone 2 g IV q24H (first 4 to 6 weeks) + Rifampicin 600-900 mg PO q24H + Co-trimoxazole 960 mg PO q12H
Bartonella spp.	Doxycycline 100 mg PO q12H + Rifampicin 300 mg PO q12H (first 6 weeks)	Doxycycline 100 mg PO q12H + Gentamicin 3 mg/kg IV q24H for 2 weeks	• Duration for Doxycycline: • 6 weeks, if valve surgery is performed. • 12 weeks, if valve surgery is **NOT** performed.
Coxiella burnetti	Doxycycline 100 mg PO		• Duration of treatment: 18 months.

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	q12H + Hydroxychloroquine 200 mg PO q8H		
Non-*HACEK* Gram-negative			
Native and Prosthetic Valves			
Pseudomonas aeruginosa *Escherichia coli* *Salmonella spp.*	Ceftazidime 2 g IV q8H **OR** Cefepime 2 g IV q8H + Gentamicin 5-7 mg/kg IV q24H **OR** Amikacin 15 mg/kg IV q24H		• The choice of antimicrobials for these microorganisms depends on antimicrobial susceptibility pattern. • Medical therapy often needs to be combined with cardiac surgery. • Duration: 6 weeks.
Fungal			
Native and Prosthetic Valves			

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Candida spp.	Amphotericin B Deoxycholate 0.6-1 mg/kg IV q24H **OR** Amphotericin B Lipid Complex 3-5 mg/kg IV q24H ± Flucytosine* 25 mg/kg PO q6H	**If unable to tolerate Amphotericin B:** Micafungin 150 mg IV q24H	• Adjust antifungal according to culture and sensitivity results. • Valve replacement is mandatory. • Duration: 6 weeks after replacement or longer in patients with perivalvular abscess. • **Step down therapy:** Fluconazole 400-800 mg (6-12 mg/kg) PO q24H for susceptible microorganism in stable patients with negative blood cultures (clearance of *Candida* spp. from blood stream). • *For synergistic effect, it causes dose related marrow toxicity. Avoid using it in patients with renal failure. • For patients who cannot undergo valve replacement, long-term suppression with Fluconazole 400-800 mg (6-12 mg/kg) PO q24H is recommended. • For prosthetic valve endocarditis, the recommendations above apply, and suppressive therapy should be lifelong if valve replacement is not possible.
Pericarditis (Purulent)			
Staphylococcus aureus *Streptococcus pneumoniae*	Vancomycin 20 mg/kg (max 2 g) IV single dose,	Vancomycin 20 mg/kg (max 2 g) IV single dose,	• Source control (pericardiocentesis) is crucial.

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Group A Streptococcus Enterobacteriaceae In post-operation, *Staphylococcus aureus* and fungi are more common. Need to consider *Mycobacterium tuberculosis.*	then 15 mg/kg (max 1 g) IV q8-12H + Cefepime 2 g IV q8H	then 15 mg/kg (max 1 g) IV q8-12H + Ceftriaxone 2 g IV q24H	• Duration: 2 - 4 weeks if adequate source control. • Intravenous therapy should be continued until fever and clinical signs of infection has resolved and until white cell count is normal, may consider oral antibiotic step down according to culture and sensitivity, if adequate source control and good oral bioavailability option. • Fluconazole 200-400 mg IV q24H may be added if patients are severely immunosuppressed or had recent ICU care or had recent broad-spectrum antibiotics. • Consider carbapenem if ESBL or MDRO is suspected.
***Streptococcal* Pharyngitis**			
No history of rheumatic fever	Phenoxymethylpenicillin 500 mg PO q8H **OR** Amoxicillin 500 mg PO q8H	**Penicillin Allergy:** Please refer to Pharmacy or ID.	• Duration of treatment: 10 days. • Household contacts should be screened and if confirmed GAS should be given a full course of antibiotic therapy.
History of rheumatic fever **(NOT receiving antibiotic**	Phenoxymethylpenicillin 500 mg PO q8H **OR**		

CHAPTER 3.2	CARDIOVASCULAR SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
prophylaxis)	Amoxicillin 500 mg PO q8H **OR** Benzathine Penicillin IM • **If ≤27 kg:** 600,000 units IM single dose • **If >27 kg:** 1.2 million units IM single dose		
Secondary prophylaxis	Benzathine Penicillin IM • **If ≤27 kg:** 600,000 units IM single dose (every 28 days) • **If >27 kg:** 1.2 million units IM single dose (every 28 days) **OR** Phenoxymethylpenicillin 500 mg PO q12H*	**Penicillin Allergy:** Azithromycin 6 mg/kg (max 250 mg) PO q24H	• Secondary prophylaxis is for patients who have had an attack of ARF, develop subsequent group A Streptococcus (GAS) pharyngitis, who are at high risk for a recurrent attack of ARF. • Duration: • **Rheumatic fever with carditis and residual heart disease (persistent valvular disease):** 10 years or until 40 years of age (whichever is longer). • **Rheumatic fever with carditis but no residual heart disease (no valvular disease):** 10 years or until 21 years of age (whichever is longer). • **Rheumatic fever without carditis:** 5 years or until 21

Infections and Likely Organisms	Suggested Treatment		Notes
	Preferred	Alternative	
			years of age (whichever is longer). • *Preferred in severe symptomatic RHD, bleeding from IM injection, low risk of recurrence
Cardiovascular implantable electronic device (CIED) infections			
S. aureus *S. epidermidis* Aerobic *gram-negative bacilli* *Candida spp.*	Ceftazidime 2 g IV q8H **OR** Cefepime 2 g IV q8H + Vancomycin 15 mg/kg (max 1 g) IV q8-12H ± Gentamicin 3 mg/kg IV q24H		• Consider adding broader Gram-negative coverage with Piperacillin/Tazobactam or carbapenem in patients with haemodynamic instability. • Duration: • Pocket or subcutaneous infection: 10 - 14 days. • Lead-associated endocarditis: up to 4 - 6 weeks. • Device removal is recommended. • Obtain blood culture from the wound, drive line, device pocket ± pump cultures, and to target antibiotics accordingly once culture is available.